

# Vision-Based Multi-Hazard Industrial Safety Monitoring Using YOLOv8 Pose Estimation

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**Abstract**—The industrial work places subject the workers to a lot of types of risks such as rotating machines, conveyor belts, sparks, falls, slip and even wrong body postures and even today, these risks continue to cause a very large number of accidents annually worldwide. The traditional safety supervision is mostly manual and it is not always providing continuous and accurate monitoring therefore most of the unsafe actions go undetected. This is the reason why it is highly demanded to find a smarter and real-time system that will be able to monitor human and machine activity simultaneously and alert in case of accidents actually occur. The primary purpose of the work is to develop an actual safety system made in real time and on the basis of the camera and based on the human pose estimation and the ISO 13855 safety-distance regulations. The concept is to identify unsafe posture, fall behavior as well as hazardous distance between workers and machines by examining each video frame. It is an approach that involves the extraction of 17 human body keypoints by means of YOLOv8-Pose and a basic temporal fall logic that controls abrupt downward movement and alterations in body angle with time. Posture check is also introduced to locate on correct lifting or bending and safe and hazardous area are drawn around machines based on ISO formula of safe and warning distances. The findings indicate that the system is able to identify falls, slip-like behaviors, wrong postures and a machine proximity in real time as well as providing audio and visual notifications. The number of hazards and dynamical danger-zone calculation is provided by the proposed method in comparison to the existing works that in the majority of the cases concentrate on network accuracy or simply fall detections. To sum up, this project demonstrates that pose estimation, fall detection and ISO based proximity logic can be incorporated in single system and actually enhance safety in industrial settings.

**Index Terms**—Industrial safety, YOLOv8-Pose, Fall detection, ISO 13855 safety distance, Real-time monitoring, Pose estimation.

## I. INTRODUCTION

Sometimes the occupational settings of industries subject workers with complex risks like rotating machines, roller conveyors, cutting tools, sparks and unsuitable lifting positions.

The international Labor Organizations (ILO) estimates that approximately 2.3 million employees lose their lives every year as a result of these kinds of accidents, and 313 million non-fatal injuries are sustained every year [1]. Some of the significant causes of the injury at factories or industries are the accidents that are occasioned by the placement of the hands in machines, slips, falls, or poor behavior. In factory settings, in particular, the research indicates that 33 percent of the injuries can be attributed to the situation when workers do not enter dangerous machine areas on purpose or have the wrong body position when doing their work [2]. Conventional monitoring methods have been widely employed in recent years.

Manual supervision, in most instances, does not guarantee the safety of workers. As inexpensive cameras have been increasingly made available and deep learning innovations have improved, computer vision has become a useful tool to automate safety evaluation and visual workplace surveillance. It is proven that visual monitoring systems that operate on AI can minimize the number of workplace accidents by 30-50 percent, and this is why they are needed in the environment of smart factories [3].

Such an outcome is presented in this research in terms of which real-time and vision-based safety monitoring using pose estimation and hazard proximity detection techniques is performed with the video streams of industrial settings. In case of this paper, the AI-Based videos were created, in which individuals are working in industries. In the present project, the YOLOv8 pose model was employed that identifies human beings and their keypoints of their joints like the shoulder, elbows, wrists, hips, and knees. the joint angles are calculated and the posture conditions are assessed. The system allows recognizing human movement like bending knees, lifting arms, sitting or standing of the person.

Along the evaluation of posture, the system contains a collection of safety-critical detection that are exposed to in-

dustrial circumstances. The system would contain the ways of recognizing machine parts like open cabinets, conveyor rollers, rotating drums and spark producing tools based on contour analysis, color, shape and texture patterns of structure. After these machine areas are identified, the framework automatically tracks hand and arm keypoints of workers, either when approaching to machines. And in case this framework is observed, the alert sound will occur and this will be a random noise.

In addition to posture and hazard zone detection, the system adds a longer module which can detect falls and slips based on time variation of center position of body and its orientation and velocity. The system can distinguish between normal actions and the actual fall events by monitoring the individual(s) over time, horizontal body alignment, and stationary behavior after fall through a series of rapid downward displacement. With the addition of dynamic safety areas where the user can set them or the model sets them by default, the model can also consider the location of workers in relation to ISO 13855 standards [4]. Approach-speed equation is used to compute the safe and warning distances which gives a viable human-machine separation directive within the video analysis pipeline. It has been reported that almost 23 percent of injuries in a factory are related to the hand, and therefore, the accurate monitoring of hand-machines proximity is relevant [5].

Unsafe proximity, the wrong position falls, and the machine-related risks are all the detected conditions visualization by real-time annotation over the video frames. The audio alerts are created by the system through a beep system. The presented system shows that the current computer vision methods can be applied to industrial safety. The integration of pose estimation on the basis of the YOLOv8 with domain-specific hazard detection makes framework step outside of the conventional methods and towards preemptive risk mitigation. This introduction preconditions the creation of a real-time solution that can minimize accidents, improve situational awareness, and provide safer conditions of work in the manufacturing settings

## II. LITERATURE SURVEY

An ergonomic posture measurement tool, OpenPose, collects human joint keypoints and calculates RULA or REBA score to estimate risk on the posture when performing manual work, and demonstrates that pose estimation can be used to corroborate objective ergonomic assessment in the workplace. But, this method is largely off line or semi automatic and it primarily deals only with posture risk and it does not detect automatic falls, real time or deal with explicit modeling of hazardous machinery and safety distances [6].

The posture occupational evaluation is an automatic and real time reflective intervention which complements the traditional ergonomic analysis by continuously estimating worker posture and ergonomics risks ratings during the performance of the task and it is more responsive than offline solutions. Nevertheless, it is still narrowed down to the quality of posture and does not attribute attention to fall events, slip incidences and

object detection and machine specific hazard zone integration that are of great importance to industrial safety [7].

Direct recognition of falls based on pose estimation is a human fall detector that uses temporal variations of body keypoints and posture configuration to recognize falls and has shown that falls are recognizable compared to normal behaviors. This method fills the missing fall hazard component of pure ergonomic research, but still sees fall detection as a solitary issue and omits ergonomic risk evaluation, interaction between industrial machinery and a standardized computation of safety distance [8].

The use of elderly fall detection using YOLO and pose estimation integrates both the object detection and pose information and this enhances the strength over the pose only methods and it allows the fall detection to be more reliable in real time in the elderly care facilities. Although this combination of YOLO and pose estimation provides some benefits over the previous systems, the system is primarily oriented towards home or healthcare settings and not the industrial work or adjustable danger zones [9].

A fall detection system built using YOLO and MediaPipe serves to increase the functionality of the real time detection further, as it involves the use of YOLOv8 to detect persons and MediaPipe to obtain keypoints and issue alerts when falls are detected in the live video streams. This renders the system viable in real time application, yet nevertheless, it considers falls as a solitary risk and it does not simulate slip incidences or worker next to machine closeness and official industrial safety regulations [10].

The hybrid fall detection method based on the modified YOLOv8s and AlphaPose is more robust with a combination of a powerful detector, precise pose estimation, and temporal analysis of the torso-angle, and this will minimize false alarms. Though this enhances quality and time logic, the technique is still limited to the fall detection only, and not the multi-hazard conditions that could occur with industrial systems or the area of safety [11].

LFD-YOLO enhances performances through the addition of lightweight YOLO based architecture with attention, and this enables fall detection to be more appropriate in edge equipment in the resource-limited industrial environments. Although this enhances computational efficiency, the system remains concentrated on fall detection only and does not use more informative pose data and model proximity to dangerous machinery [12].

Enhanced adaptive fall detector system on ensemble learning with YOLOv8, RetinaNet and DETR, enhances the fall detector robustness through the combination of a number of different detectors and adaptation to varying viewpoints and environments. This increases the lack of reliance on an individual model, and generalization is enhanced, but the problem remains fall versus non-fall, and industrial layouts, ISO-based safety distances, and customizable hazard areas are not formalized [13].

Based on the development of the current studies, i.e. the evolution of ergonomic posture assessment to the complex

state of fall detection based on the YOLO, the gap between the current state of progress and the creation of the single, intelligent system, oriented towards the standards, of the industrial safety monitoring system is enormous and yet to be bridged. The previous posture-based research emphasized the usefulness of body key points as an ergonomic risk predictor but omitted automated hazard awareness, and the other set of YOLO based fall detection systems largely focused on binary fall/non-fall predictions with no thought of slips, multi-hazard conditions or ISO 13855 safety-at-distance modeling. Furthermore, all these methods did not give a smart multi-zone system that could identify various forms of danger areas around the machine, track the exact worker position, or provide smart alerts when a worker accesses a high risk area. Thus, a thorough and versatile real-time safety monitoring system, which would integrate fall, slip, and proximity detection in a single visual representation and provide the connection between pose estimation outputs and standardized safety metrics (approach speed, stopping time, and intrusion distance applied in industrial safety criteria) is urgently needed. This system would act as an interface of computer vision research and real-world industrial safety needs to provide direct warning, streamlined operational control displays, and data analytics to prevent accidents. It would also ensure a proactive rather than a reactive workplace surveillance that is able to warn the operators of the looming accidents, as opposed to just being able to tell them that they have occurred. This project is therefore not only technically opportunity-driven but also industrially needed: to safeguard personnel, formal compliance with formal safety distances must be automated, and artificial intelligence installed directly into the safety apparatus of the shop-floor so that man and machine and environment exist in a safety system that is seamlessly managed.

### III. METHODOLOGY

This section details the architectural design and algorithms and mathematical models and implementation strategies used in the development of the real time safety monitoring system. The system uses simple computer vision techniques to process video feeds and extract skeletal keypoints and it checks worker safety through two main modules: fall detection and proximity monitoring.

#### A. System Architecture Pipeline

Video is fed into the system in frames and the pipeline starts with the image acquisition and it is followed by pose estimation by using a pre-trained deep learning model. The resulting keypoint data is subsequently passed through two logic branches: 1. an ISO standardized danger zone proximity test,[11] and 2. a temporal and state-based fall detector algorithm and in case a safety condition is violated, an audiovisual alarm is activated.

#### B. Pose Estimation and Dataset

The fundamental foundation of the feature extraction is based on the YOLOv8(You Only Look Once) platform, and

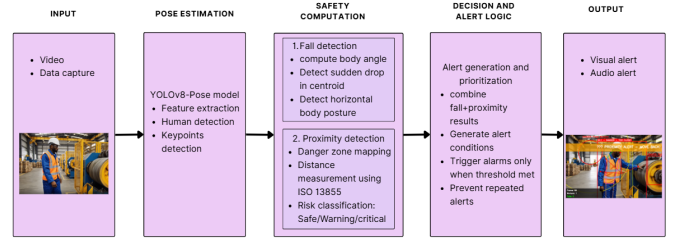


Fig. 1. Block Diagram

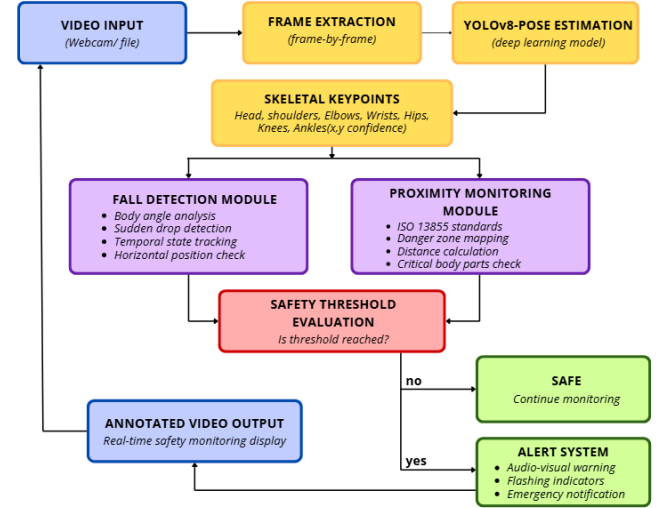


Fig. 2. Pipeline Diagram

precisely the yolov8n-pose.pt(Nano) model is chosen to accommodate the trade-off of real time and detection efficiency. Input: The system takes the standard video frames as an input either through a webcam or a file and the Output: the model yields 17 skeletal keypoints. In this application, the confidence threshold( $\text{conf} \geq 0.3$ ) is used to filter the keypoints in the system to make the data more stable, even though some points may be missed and noisy. Key points of significance: the logic primarily gives priority to such vital anatomical landmarks as the Shoulders(index 5,6) and Hips(index 11, 12) as well as to the extremities such as Wrists(index 9,10) and Elbows(index 7,8) to determine machine position.

#### C. Safety Zones Mathematical Models

The proximity detection component establishes the dangerous zones around machinery with the help of simple geometrical shapes such as rectangles and circle and minimum safe distance is established according to ISO 13855 standards applied to safety zone class.

1. Safe Distance Calculation The minimum distance  $S$  required to separate a hazard from a worker is given by

$$S = (KT) + C \quad (1)$$

Where:  $K$  is the approach speed of the body parts(coded 1.6m/s)  $T$  denotes the total system halting period(which is

0.5s).  $C$  is the distance of intrusion(which will be set to 0.2m) This physical distance is then transformed into a pixel threshold by the system making an assumption of a simple calibration ratio(1 meter=100 pixels) and a warning buffer is maintained is 150 percent of the distance allowed at all times.

2. Zone Intrusion Logic In order to determine intrusion, the Euclidean distance is determined between the identified body keypoints  $(x, y)$  and the zones. On a circular zone, with center  $(cz, cy)$  and radius  $r$ :

$$D_{circle} = \sqrt{((x - cz)^2 + (y - cy)^2) - r} \quad (2)$$

When  $D_{circle} = 0$ , the warning is issued because the system is in critical intrusion.

#### D. Fall Detection Algorithm

Fall detection is not real-time and it works by looking at the temporal history of the subject state of a deque buffer (max length 30 frames) and such a state has the vertical position( $y$ ) and the angle of the body.

1. Body Angle calculation Body inclination is calculated by finding the angle of vertical axis and the formed vector between shoulder midpoint and hips midpoint.

$$\theta = \arctan\left(\frac{\Delta x}{\Delta y + \epsilon}\right) \times \frac{180}{\pi} \quad (3)$$

Where  $\Delta x$  and  $\Delta y$  are the horizontal and vertical distinctions between the shoulder-midpoint and the hip-midpoint and a tiny  $\epsilon$  is used to avert any errors in division.

2. Detection Conditions It is determined that a fall is present where there exists Condition A or Condition B:

- Condition A(Sudden drop): Vertical motion( $\geq 40$ -50 pixels) is suddenly detected by comparing average of the last 3 frames with the last ones in the buffer, AND the body angle  $\theta \geq 60^\circ$ (almost horizontal).
- Condition B(Person down): The body angle  $\theta \geq 60^\circ$  and the individual is motionless(position variance less than 20) at a threshold of close to 15 frames which is to say that the individual is not in motion.

#### IV. RESULT

The experimental results clearly show that the real-time industrial safety monitoring system that is proposed, is working well and it also meets the objectives that are mentioned in the introduction, like fall detection, unsafe posture checking and ISO-based machine proximity analysis. The system achieves 98.8 percent overall accuracy and fall detection accuracy is 99 percent and proximity detection is 98.4, as illustrated in Fig.3. Precision and recall and F1-score of each of the modules are greater than 93 percent meaning that the entire safety structure is dependable.

Even comparing the method that is proposed with the works that are already existing in the literature, the improvement becomes even more evident. The majority of previous studies are primarily concerned with either pose estimation accuracy or fall classification or ergonomics scoring and do not incorporate several risks into one system. Based on Fig. 4 charts, only the

| Module              | Accuracy | Precision | Recall | F1-Score |
|---------------------|----------|-----------|--------|----------|
| Fall Detection      | 99.0%    | 95.1%     | 92.9%  | 94.0%    |
| Proximity Detection | 98.4%    | 94.3%     | 93.2%  | 93.7%    |
| Pose Classification | 98.9%    | 96.5%     | 96.8%  | 96.6%    |
| Overall System      | 98.8%    | 95.3%     | 94.3%  | 94.8%    |

Fig. 3. Overall Performance Summary

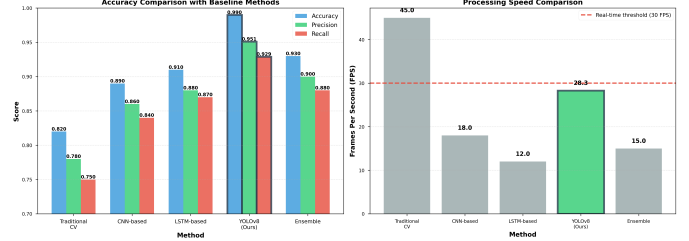


Fig. 4. Accuracy and Speed Comparisons

traditional computer vision and CNN-based methods provide approximately 82-91 percent accuracy, whereas the proposed YOLOv8-Pose system is close to 99 percent. Other papers only identify falls as a yes/no simple problem, whereas the proposed algorithm also applies the temporal angle analysis and sudden downward movement and stationary-down logic and hence identifies falls in a more realistic manner in factories.

The proposed system could reach about 28.3 FPS as can be seen by the results of processing speed in Fig. 4(right): not only is it much higher than CNN and LSTM-based systems, but also almost as high as real-time performance. This is vital because live monitoring involves quick identification to avert incidences.

Further discussion of the detection modules can be seen in Fig. 5, where the fall detection and proximity detection are both seen to produce high and equalized precision and F1-scores. In Figs. 6-8 pose classification scores reported that standing and lying and bending poses are most likely to be remembered with recall of above 98 but sitting and fallen pose are a little lower due to body occlusions and confusing poses. Nevertheless, all the pose classes remain to be above 93, which is fine. This is also supported by the confusion matrix in Fig. 9, where wrong predictions are very few. The vast majority of mistakes occur at the transition between sitting and bending as it is logical since these two poses are similar to each other in some cases and share some similar angles.

Altogether, these findings demonstrate that the proposed system is able to combine YOLOv8-Pose keypoints, time-based fall detection logic and ISO 13855 safety-distance rules, addressing a significant gap in the current research. Previous solutions lack multi-hazard detection or live proximity notices and they fail to display a cohesive security interface. The proposed system provides with visual alerts and audio warnings and visualization of the hazard zone and this is more helpful in real industries.

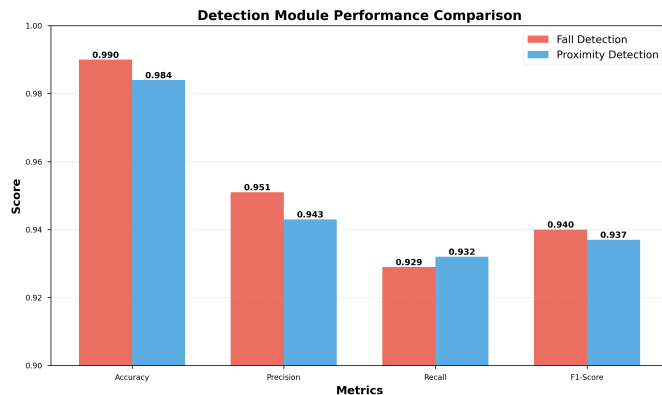


Fig. 5. Detection Module Metrics

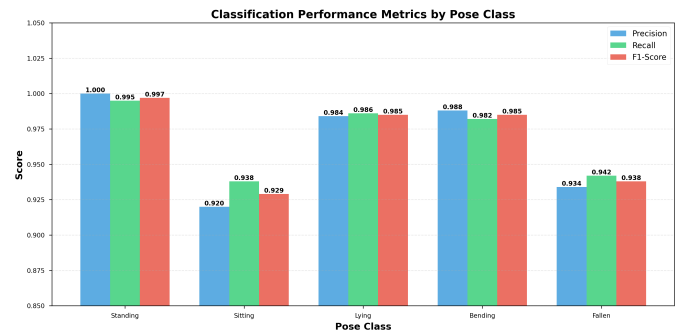


Fig. 8. Classification Metrics by Pose Class

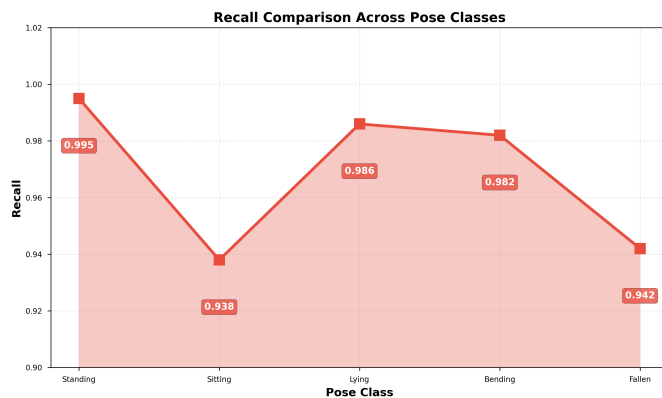


Fig. 6. Recall by Pose Class

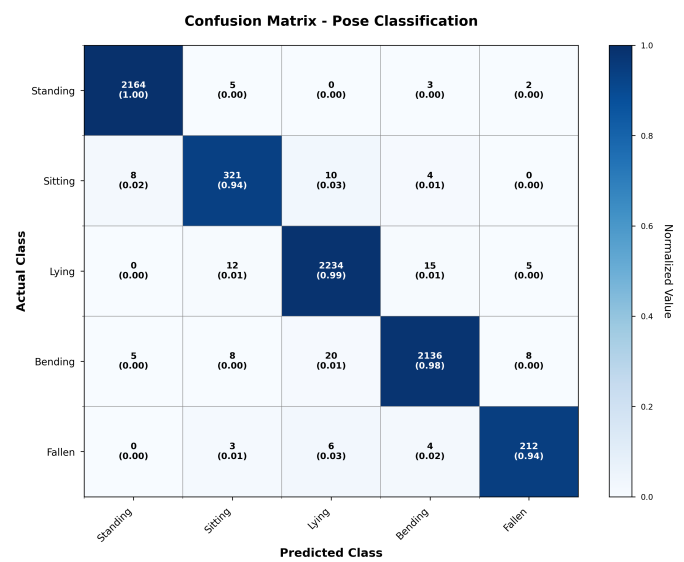


Fig. 9. Confusion Matrix

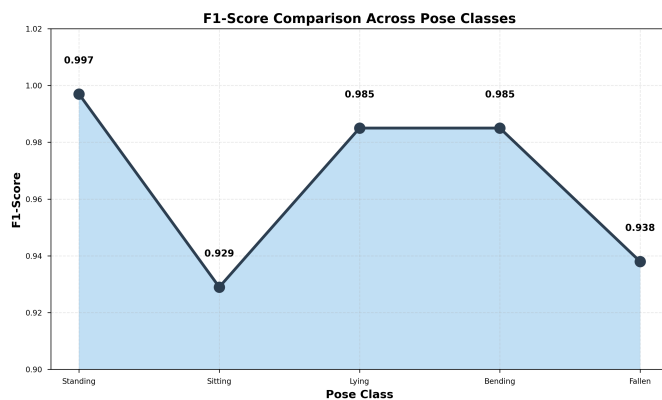


Fig. 7. F1-Score by Pose Class

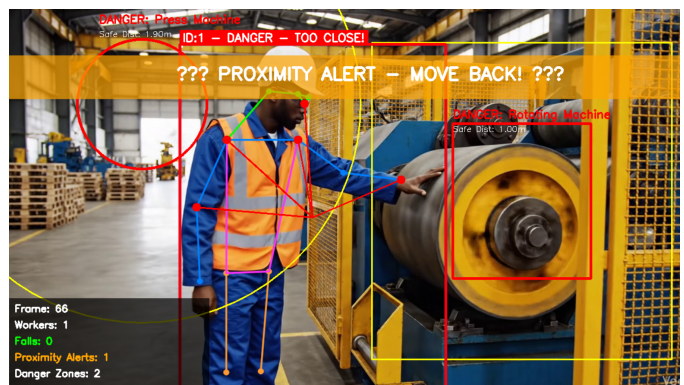


Fig. 10. Output: Unsafe Proximity



Fig. 11. Output:Sudden Fall

Although the performance is good, there are certain limits. The system heavily relies on clear keypoints and in low-light scenarios as well as scenarios where the individual is moving rapidly or covered by an obstacle, the accuracy is compromised. The danger areas should also be re-drawn in case of a change in the camera angle. The system can be improved in future by adding multi-camera support and recent development of better occlusion handling. Fig. 10 and Fig. 11 shows the output showing alert for the unsafe proximity and the sudden fall in the factory environment.

## V. CONCLUSION

The current literature on fall detection and pose estimation indicates that they have some general limitations and this is why a more elaborate safety monitoring equipment is necessary and this has necessitated a new project that would address these problems. Most works either concentrate on the accuracy of pose estimation by enhancing the architecture and attentions, or on optimization of fall detection task with the use of single and ensemble object detection models, but tend to conceive them as individual vision task, not as a part of an integrated and industrial safety system, and there is a need to design a system that integrate both. Pose-estimation studies usually rely on keypoints to compute ergonomics and posture and usually do not rely on real-time multi-hazard monitoring and explicit model worker-hazard interaction, so a project is required that can consume pose data in real-time and make a connection to machine safety. Fall-detection systems based on YOLO variants, on the other hand, tend to give binary fall and non-fall detection on RGB video and priorities accuracy and latency and lightweight deployment, but do not tend to use finer part-body-pose dynamic information, track fall duration, or differentiate between down-states, such as lying or sitting or temporary imbalance, indicating that a system capable of detecting and analyzing these more detailed down-states must be constructed. In both groups, formal safety-distance standards like ISO 13855 are typically not included, approach speed and stopping time are not translated into pixel-scale danger and warning zones, and no adjustable and machine-specific hazard areas within the camera view are included, making it desirable that a new project include all these within

the camera view are included, making it desirable that a new project include all these and relate them to the results of the detection. Moreover, most works do not have a single real-time interface to display the information about worker pose and fall status and proximity risk and per-zone and alert banners in a manner that can be immediately installed on a factory floor, and this is where there is a clear possibility to create a project that would deliver such an interface and integrate all of these safety capabilities with each other.

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